Real-Time Detection of Personal Protective Equipment Violations for Construction Workers Using Semisupervised Learning and Video Clips

Qihua Chen[1]; Danbing Long[2]; Siqi Wang[3]; Qirong Chen[4]; and Beifei Yuan, S.M.ASCE[5]

Abstract: The collection of personal protective equipment (PPE) violation data is crucial for assessing safety risks in behavior-based safety (BBS) management on construction sites. Owing to the limitations of manual inspection methods, many studies have employed computer vision–based methods for PPE violation detection. However, limitations exist in implementing this in actual construction projects due to the costs associated with the acquisition and labeling of a large number of images, and the accuracy and efficiency of recording PPE violation data. Therefore, this study introduces semisupervised object detection (SS-OD) and data augmentation for training PPE detection models to reduce the use of labeled data without reducing the performance, and proposes a framework for recording the PPE violation data by extracting PPE violation videos from real-time surveillance footage by considering the role of the worker and their position, thereby enhancing practical construction management and serving as input for BBS-based safety risk assessments on construction sites. The results show that (1) the labeled data demand for training PPE detection models can be reduced through semisupervised learning, image augmentation, and transfer learning, without reducing the performance of PPE detection, thereby reducing the cost of application on actual construction sites; (2) SS-OD methods are better equipped to handle changes in construction scenarios by making full use of unlabeled data, and thus are suitable for construction scenarios; and (3) recording PPE violation data using the video clip method in a real construction project achieved an average precision of 91.76% and an average $F1$ score of 92.79%. Using the PPE detection model trained with SS-OD effectively records PPE violation data for BBS-based safety risk assessment. This study significantly enhances the efficiency of real construction site PPE violation inspections and provides a valuable method for the automated and real-time collection of violation data in BBS-based management.
DOI: 10.1061/JCEMD4.COENG-15310.

Practical Applications: Computer vision–based personal protective equipment violation detection methods from past studies have been difficult to implement widely in real projects due to cost and effectiveness limitations. This study effectively improves upon these limitations using semisupervised learning and video clipping methods. Experimental results demonstrate that the object detection model trained with the semisupervised learning method not only reduces the need for labeled data in personal protective equipment detection model training but also exhibits improved generalization capabilities in complex and dynamic construction scenarios. Using the personal protective equipment violation video clipping method proposed in this study allows for efficient editing of violation video clips, facilitating rapid identification and documentation of construction workers' personal protective equipment violation behaviors. This enhances the safety culture at construction sites and aids construction managers in developing safety management measures and educational plans. The real-time-collected personal protective equipment violation data can be utilized further as input for a behavior-based safety risk assessment model, enhancing the automation of unsafe behavior data collection.

Author keywords: Personal protective equipment (PPE); Behavior-based safety (BBS); Construction safety management; Semisupervised object detection; Violation video clip.

Introduction

The behavior-based safety (BBS) method for construction safety management is an important way to improve worker safety performance and avoid accidents (Chen et al. 2023b; Hu et al. 2023; Vinodkumar and Bhasi 2010). Lee et al. (2019) and Chen et al. (2023b) proposed BBS methods for comprehensive and dynamic risk analysis at construction sites. However, PPE violations, which are critical for risk analysis and are prevalent at construction sites, were recorded only sporadically by site managers (Xu and Wang 2023), and this approach was considered to be time-consuming, inefficient, and error-prone (Awolusi and Marks 2017; Torabi et al. 2022). Because construction is a labor-intensive industry (Fang et al. 2020), it is impractical for a limited number of safety managers to monitor each worker's PPE usage constantly (Vukicevic et al. 2022). Fortunately, construction sites have a substantial number of high-definition cameras installed (Torabi et al. 2022), and with

[1]Master's Candidate, School of Civil Engineering, Southwest Jiaotong Univ., Chengdu 610031, China. ORCID: https://orcid.org/0000-0003-4701-4533. Email: cqh@my.swjtu.edu.cn

[2]Lecturer, School of Civil Engineering, Southwest Jiaotong Univ., Chengdu, Sichuan 610031, China (corresponding author). Email: lornalong@swjtu.edu.cn

[3]Master's Candidate, School of Civil Engineering, Southwest Jiaotong Univ., Chengdu 610031, China. Email: Waskiiiii@outlook.com

[4]Master's Candidate, School of Civil Engineering, Chongqing Univ., Chongqing 400045, China. Email: cqr@stu.cqu.edu.cn

[5]Ph.D. Candidate, School of Management Science and Real Estate, Chongqing Univ., Chongqing 400045, China. Email: Yuanbeifei@stu.cqu.edu.cn

Note. This manuscript was submitted on March 9, 2024; approved on September 3, 2024; published online on December 27, 2024. Discussion period open until May 27, 2025; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Construction Engineering and Management*, © ASCE, ISSN 0733-9364.

the development of computer vision technology, these videos can realize automatic PPE detection in nonintrusive ways (Kim et al. 2018). For example, in recent years, researchers have been working on using computer vision techniques to detect the wearing of helmets and safety vests on construction sites (Cheng et al. 2022; Nath et al. 2020; Shahin et al. 2023). Therefore, the vision-based approach holds great potential for PPE violation monitoring on construction sites.

According to previous research (Alateeq et al. 2023; Iannizzotto et al. 2021; Lee et al. 2023; Udatewar et al. 2021), the workflow of common computer vision–based PPE detection involves (1) collecting data sets and training the PPE detection model, (2) using trained detection models to detect PPE on construction sites, and (3) determining whether PPE violations exist. However, costs associated with image acquisition and labeling, accuracy, and efficiency of recording PPE violation data hinder the application of these methods in actual construction projects. In the first step, prevailing research on PPE detection models predominantly utilizes supervised object detection (S-OD) techniques. However, S-OD methodologies necessitate extensive labeled data sets to train highly precise detection models (Kim and Chi 2021). Large-scale data collection and labeling increase costs, resulting in difficult application in practical construction projects. Wan et al. (2023) called for semisupervised object detection (SS-OD) methods to reduce the use of labeled data in the PPE detection model training. SS-OD uses a small number of labeled data and a large number of unlabeled data to jointly train the detection network. This greatly reduces the work of labeling image data while maintaining accuracy by making full use of unlabeled data (Xiang et al. 2023). Several studies have investigated the application of SS-OD methods in construction field object detection tasks, and have achieved better results than with the S-OD method, for example, in detecting defects in facades (Guo et al. 2021) and detecting construction machinery (Xiao et al. 2021b). These studies used a two-stage object detection algorithm as the detection network. For example, Xiao et al. (2021a) used Faster R-CNN as the teacher and student models in SS-OD methods. However, although two-stage object detection algorithms are more accurate, they are slow due to the extensive computational demands of their network structure parameters. Therefore, they fail to meet the requirements for real-time detection in application areas (Shahin et al. 2023).

Another existing limitation, in Step 3, hinders the use of vision-based PPE detection methods in construction safety management because the results are not directly usable for further construction safety risk assessments. This is reflected mainly in three aspects:

1. Lack of flexibility. Although previous studies could detect whether workers in images wore PPE, they often ignored variations in appropriate PPE use across different roles and contexts. Moreover, they did not consider variations in PPE use due to workers' locations (Wan et al. 2023).
2. Limitations in the ability of PPE detection models. The long detection distance at construction sites can cause PPE either to not be detected or to be incorrectly detected, leading to erroneous determinations. For example, the experimental results of Zhafran et al. (2019) using Faster R-CNN to detect PPE at different distances showed that the detection accuracy decreased drastically at 5 m.
3. Lack of processing of surveillance videos and generation of violation data for risk assessment. Previous studies were able to recognize individual images only without considering the continuous relationship between video frames. Additionally, there was no further clipping of the violated video clips for easy inspection and analysis by construction safety managers, nor for recording data for construction safety risk assessment.

Xu and Wang (2020) proposed a safety warning mechanism based on a computer vision–based data collection method and BBS, including a comprehensive risk assessment model and a safety warning system. However, the data are recorded in units of each video frame, which leads to error detection and subsequent system alerts, making the proposed method unreliable for use in real projects. Several studies have improved the management efficiency by clipping construction videos. For example, Xiao et al. (2021a) automatically generated video highlights from construction videos based on detecting construction machinery, which effectively reduced the video storage space. Ham and Kamari (2019) retrieved images that were valuable for documenting the current status of construction sites from video clips captured by drones. However, video editing of PPE violations differs from the aforementioned tasks due to the complexity and diversity of workers on construction sites, e.g., workers with different roles and locations often require different PPE.

This study proposes a framework for PPE violation detection to reduce the cost of vision-based PPE violation detection and to efficiently collect PPE violation data in practical construction projects for further using in safety risk assessment. Specifically, a SS-OD method based on YOLO and data augmentation is utilized to train the PPE detection model, and its effectiveness in reducing the use of labeled data was assessed. Furthermore, this study proposes a PPE violation video clip method for extracting the corresponding violation videos and records from construction surveillance footage. The method begins by defining the PPE rules according to different helmet colors and camera IDs. Subsequently, it assesses whether the worker in the current image complies with these PPE rules. Finally, based on the PPE violation sequence, video clips showcasing PPE violations are extracted from the surveillance footage. This study also created a 13-class construction PPE data set (CPPED), based on existing publicly available data sets, to verify the proposed method.

Literature Review

Computer Vision–Based Construction PPE Detection

Construction workers' PPE violations are important indicators of unsafe behavior in BBS-based safety management. Numerous studies have concluded that most injuries and deaths could be prevented if workers wore appropriate PPE (Al-Bayati et al. 2023; Nath et al. 2020; Rasouli et al. 2024; Wang et al. 2023c; Yuan et al. 2022). For example, a significant number of deaths caused by slips and trips, and those resulting from being struck by falling objects, could be reduced (Fang et al. 2018b). Common PPE in construction mainly includes helmets (of different colors), safety vests, protective gloves, safety shoes, protective masks, and safety glasses (Fig. 1). For example, helmets play a crucial role in preventing accidents and mitigating injuries. Firstly, helmets are used to reduce the extent of injuries in accidents, such as those caused by object strikes, people falling from heights, or other types of falls, due to their ability to absorb and disperse impact forces (Hayat and Morgado-Dias 2022; Wang et al. 2021). Previous studies have shown that helmets can reduce the risk of brain injury from concrete block impacts by as much as 95% (Suderman et al. 2014). Secondly, the helmet's bright color enhances worker' visibility, especially in rainy conditions or low-light environments. Furthermore, the color of the safety helmet serves to differentiate among different roles, facilitating personnel management on the construction site (Wan et al. 2023; Wang et al. 2021). For example, Chinese regulations stipulate that managers wear white helmets and technicians wear blue

Fig. 1. Commonly used personal protective equipment (PPE) in construction.

helmets (General Administration of Quality Supervision 2013). Protective glasses are used to prevent bright light radiation and material from reaching the operator's eyes. A study by the US Bureau of Labor Statistics (BLS) showed that almost three-fifths of workers with eye injuries were not wearing any protective shield at the time of the accident (Ferdous and Ahsan 2022). Therefore, detecting and recording whether construction site workers appropriately wear such PPE and managing them in a focused manner can reduce safety risk.

Constructing a PPE data set with more classes is a prerequisite to detecting more categories of PPE. The GDUT-HWD (Wu et al. 2019) data set is notable as the earliest data set to include four different colors of helmets. The CHV (Wang et al. 2021) data set introduced the safety vest category and four categories for helmets (each with a different color), totaling five categories. The CHVG (Ferdous and Ahsan 2022) data set expanded upon the CHV data set by introducing the protective glasses category. Ngoc-Thoan et al. (2023) developed a data set containing six categories of PPE, including safety shoes, safety vests, protective gloves, and protective glasses. Wang et al. (2023c) created the PPE-dark data set, which focuses on three construction PPE categories specialized for low-light scenes in tunnels.

However, more consideration should be given to the parts of the body not wearing PPE, because detecting these parts will ascertain the absence of the corresponding PPE. Several studies, such as those by Gallo et al. (2022), Lo et al. (2023), and Liu et al. (2023), on construction PPE data sets, include body parts in the detection classes. For example, detection of a worker's head can help ascertain whether a helmet is worn, and the detection of a worker's hand can determine if protective gloves are in use. Despite the gradual diversification of current data categories, more categories and high-quality data sets are still necessary. Nath et al. (2020) and Li et al. (2022b) called for future research to expand the categories of PPE to promote wider utilization.

However, creating an extensive construction PPE data set, which involves the collection and annotation of large data sets, remains challenging (Ghelmani and Hammad 2023; Jiang et al. 2023; Wang et al. 2019). Furthermore, labeling images requires expertise in both deep learning and construction management, making it a time-consuming, labor-intensive, and costly process (Xiao et al. 2021b). According to Xiao and Kang (2021), the cost of annotating an image is about $0.51. Additionally, the training data sets for PPE detection models often are fixed, and those used in various construction studies frequently are limited in size (Fang et al. 2018c). Consequently, the detection models trained on these data sets may not achieve high accuracy in all types of scenes (Wang et al. 2023a; Zhai et al. 2023). Therefore, the detection accuracy can be constrained by the available labeled data.

For vision-based PPE detection, Faster R-CNN (Ren et al. 2015), YOLO (Redmon et al. 2016), and SSD (Liu et al. 2016) are the predominantly used methods (Fang et al. 2018a; Gallo et al. 2022; Wu et al. 2023). Faster R-CNN, which is a two-stage object detection method, is challenging to apply in industrial practice due to its slow detection speed (Chen et al. 2023a; Diwan et al. 2023; Ye and Chen 2023). Single-stage algorithms, which have faster detection speed, are used more widely in industry (Li et al. 2023). Both the YOLO and SSD algorithms are representative algorithms for single-stage object detection. However, YOLO has been improved significantly over time and is more accurate than other single-stage object detection methods (Ngoc-Thoan et al. 2023; Xie et al. 2018; Xu et al. 2021), and it has been the most popular detection framework for industrial applications (Ye and Chen 2023). Iannizzotto et al. (2021) compared EfficientDet (which is similar to but outperforms SSD) and YOLOv5s, and showed that the YOLOv5s model achieved both a shorter average inference time and superior detection performance. Furthermore, numerous previous studies of construction PPE detection have utilized various versions of YOLO. Delhi et al. (2020) used YOLOv3, Protik et al. (2021) used YOLOv4, Ngoc-Thoan et al. (2023) used YOLOv5, Athidhi and Smitha Vas (2023) used YOLOv7, and Wang et al. (2023c) used YOLOX to detect construction PPE. Therefore, the YOLO algorithm can be used to meet the detection speed and accuracy required for PPE detection on construction sites.

Semisupervised Object Detection

Semisupervised object detection, which combines supervised and unsupervised learning, can achieve a better balance between the two (van Engelen and Hoos 2020; Xiang et al. 2023). For example, it offers better generalization ability (Cai et al. 2022) and robustness (Xiao et al. 2021b). SS-OD has two main directions: consistency regularization, and pseudo-labeling (Li et al. 2022a; Yang et al. 2022). Consistency regularization (Sajjadi et al. 2016), also known as consistency training, assumes that randomness in neural networks (e.g., using dropout) or data augmentation transformations should not change model predictions for the same inputs. Consistency constraints can be considered at three levels: the input data set, the neural network, and the training process (Yang et al. 2023). Pseudolabeling (Lee 2013) is an efficient and simple SS-OD technique, which has the following self-training process (Meng et al. 2023; Xiao et al. 2021b): (1) train the teacher model on labeled data; (2) generate pseudolabels on unlabeled data using the teacher; (3) train the student model on labeled and pseudolabeled data; and (4) repeat the process until convergence. For example, Shim et al. (2022) proposed a method to significantly improve road defect detection performance by expanding the training images using pseudolabels. Recently, SS-OD methods have focused increasingly on the pseudolabeling framework (Ghelmani and Hammad 2023; Xu et al. 2023).

In the past, methods based on pseudolabeling for two-stage object detection (e.g., a teacher and student model using R-CNN) have produced better results. In the construction industry, Xiao et al. (2021b) used a SS-OD method based on a Faster R-CNN detection network to train a construction machinery detection model, and achieved 91% mean average precision (mAP) while requiring only 30% of the training data set. Guo et al. (2021) proposed a SS-OD method, based on a two-stage object detection network, for detecting defects in facades. Validation experiments showed that the proposed method improved model accuracy from 79.26% to 84.36% compared with traditional S-OD methods when 10% of labeled data were used in the training data set. Shi et al. (2024) trained a Faster R-CNN–based object detection model using unlabeled construction machinery images using a single-stage semisupervised learning method, and achieved an average accuracy of 81.1% with only 3% of labeled images. However, the semisupervised learning training framework used in previous studies is unsuitable for training single-stage object detection models (Xu et al. 2023), such as the YOLO series, due to the inconsistency of pseudolabels generated and the efficiency of model training.

Construction PPE Violation Detection

Surveillance videos can facilitate safety inspections (Han and Lee 2013). However, given the numerous surveillance cameras at construction sites, it is challenging for construction management. For example, construction surveillance videos contain many invalid video clips, leading to fatigue from long periods of video monitoring. Numerous studies have proposed computer vision methods for detecting PPE compliance among workers. Delhi et al. (2020) detected compliance with helmet and safety vest usage among workers using the YOLOv3 algorithm. Iannizzotto et al. (2021) proposed a near-real-time PPE detection framework for embedded platforms that combines object detection with fuzzy filtering. Cheng et al. (2022) developed a framework that combines worker reidentification and PPE classification based on computer vision techniques to monitor the compliance of worker helmet and safety vest usage. However, these studies mainly addressed partial PPE wear violations or discussed the accuracy of object detection (Alateeq et al. 2023; Ngoc-Thoan et al. 2023). Little consideration has been given to further applications in construction projects, such as clipping violation videos from construction monitoring. The time information in the video can capture changes over time, which can provide greater value for construction site surveillance videos (Wang et al. 2023b). Xiao et al. (2021a) used a vision-based approach to clip video highlights from construction surveillance, such as construction site entrance and exit vehicles. However, given the diversity of violation types on construction sites, considering only the appearance of objects in video frames is insufficient. The role and location of the worker in the actual construction and the recording of data for further use in construction safety assessment must be considered. Therefore, developing an efficient method for extracting construction PPE violation clips that meets the practical needs of construction projects and utilizes them for safety risk assessments is necessary.

Methodology

The overall framework of this study consists of two parts (Fig. 2): SS-OD, and real-time PPE violation detection. The SS-OD method, utilized to train the construction PPE detection model, is divided into three subcomponents: data processing, a supervised training stage, and a semisupervised training stage. In the data processing stage, image data from construction site acquisitions and public data sets are filtered and labeled to segregate labeled and unlabeled data. These data subsequently are enhanced using both weak and strong image-augmentation techniques. During the supervised training stage, the augmented labeled data are used to train the object detection model. The semisupervised training stage incorporates a teacher–student model framework, wherein the teacher model, initialized with results from the supervised training, processes weakly augmented unlabeled data to generate pseudolabels. The student model is trained using weakly augmented labeled data and strongly augmented unlabeled data, and the teacher model is updated synchronously during the training process using the exponential moving average (EMA) method. In the real-time PPE violation detection part, the student model detects the PPE by surveillance footage. The PPE rules, which are set by safety manager, and the PPE violation determination method then are used to determine if a violation has occurred. Then the violation videos are clipped automatically from the surveillance video based on the information from a continuous violation video frame. Finally, the extracted violation videos and their corresponding PPE violation types are saved for further action in safety management.

Data Set Process

Based on the existing open-source construction PPE data set and the current image data collected from a real construction site, the original data, denoted D_{raw}, are used for model training (Fig. 2). The raw images often are blurred or small. Therefore, all the collected raw image data need to be filtered manually to select high-quality images. Only a small portion of the image data is selected for labeling. Finally, the labeled and unlabeled data sets are obtained. The labeled data set is denoted $D_L = \{(x_i, y_i)\}_{i=1}^{N}$, where $x_i (x_i \in D_{\text{raw}})$ denotes the labeled image, y_i denotes the corresponding label, and N denotes the number of labeled images. The remaining samples are unlabeled images. These unlabeled image data are denoted $D_{UL} = \{x_i\}_{i=1}^{M}$, where $x_i (x_i \in D_{\text{raw}})$ denotes the unlabeled image, and M denotes the number of unlabeled images.

Data augmentation is a crucial step in data processing. Data augmentation allows for the acquisition of a large volume of unique data, thereby improving the model's generalization capability. In this study, both weak and strong data augmentations are applied to produce the final augmented data set. The Mosaic method (Bochkovskiy et al. 2020) is utilized for weak data augmentation, which involves randomly flipping, scaling, and color-transforming four images before stitching them together into one image. This method enhances the contextual information of the images, reduces batch-size requirements, and increases learning efficiency. Mosaic also significantly broadens the context of detected objects, particularly the random scaling which introduces many smaller objects, thereby enhancing the network's robustness. The weakly augmented labeled data are denoted $D_L^w = \{(x_i^w, y_i^w)\}_{i=1}^{Q}$, where Q denotes the number of weak augmentations applied to the labeled images. The weakly augmented unlabeled data are denoted $D_{UL}^w = \{x_i^w\}_{i=1}^{P}$, where P denotes the number of weak augmentations of the unlabeled images. Building upon weak data augmentation, strong data augmentation employs the cutout method (DeVries and Taylor 2017). In this method, a randomly selected point in the image becomes the center of a fixed-size square zero-mask. Cutout enables the neural network to leverage the entire image's global information, rather than merely the local information represented by specific small features. The strongly augmented unlabeled data are denoted $D_{UL}^s = \{s(D_{UL}^w)\}_{i=1}^{P}$, where s denotes the strong augmentation method. Fig. 3 shows sample images generated using both strong and weak augmentation methods.

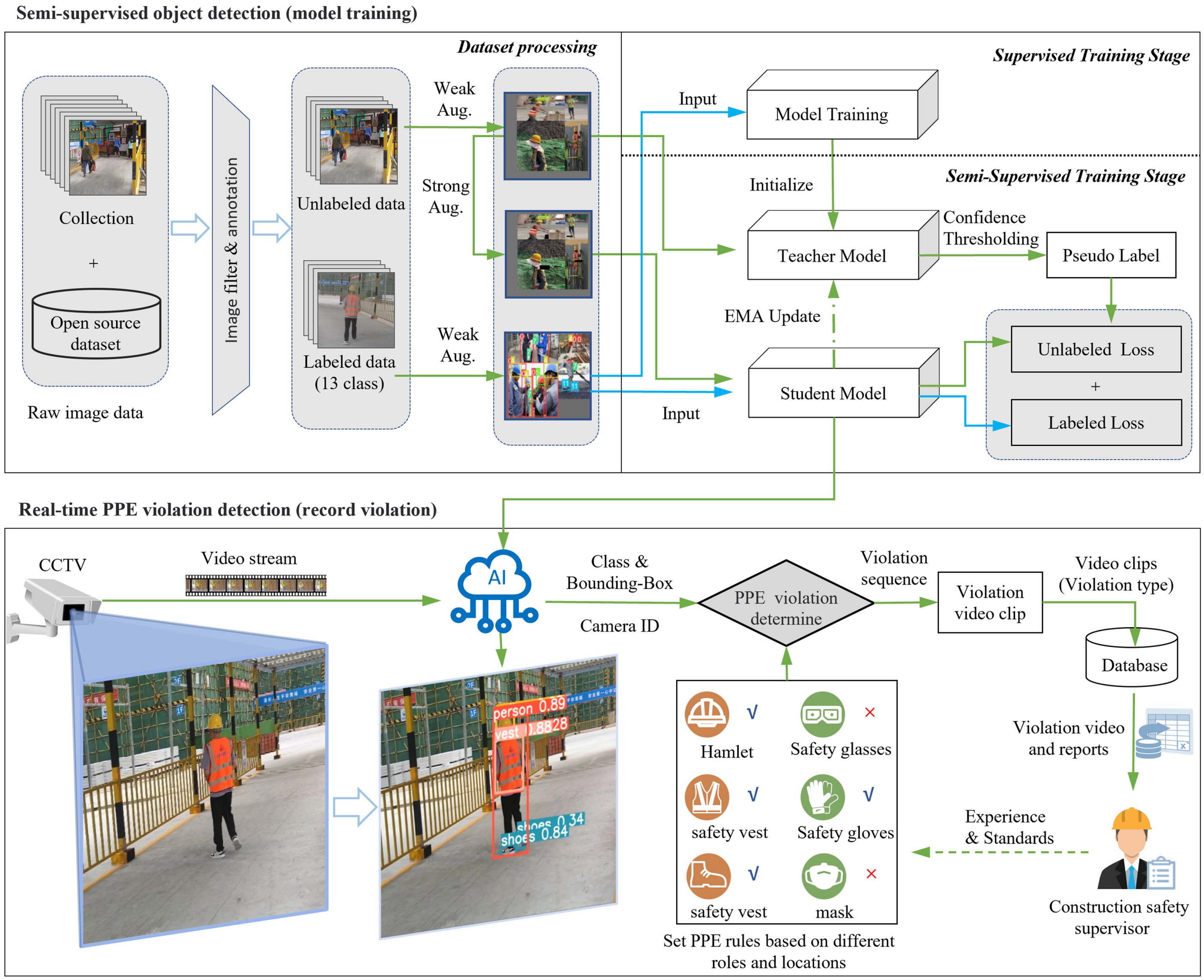

Fig. 2. Framework of SS-OD and real-time PPE violation detection. (Images by Qihua Chen.)

Fig. 3. Strong and weak augmented images: (a) weakly augmented labeled images; (b) weakly augmented unlabeled images; and (c) strongly augmented unlabeled images. (Images by Qirong Chen.)

Semisupervised Object Detection

During the supervised training stage, the object detection model is trained using weakly augmented labeled data D_L^w. The detection network utilized for S-OD in this study is YOLOv5. S-OD is used to obtain the optimal model, which subsequently initializes the teacher model for semisupervised training stage. In the semisupervised training stage, the teacher–student semisupervised training framework is used (Fig. 2). A training strategy based on single-stage object detection (Xu et al. 2023) is proposed, which reduces the inconsistency of pseudolabels in SS-OD by utilizing pseudo label assigner (PLA) and improves the efficiency in training by using epoch adaptor (EA). In this study, YOLOv5 was utilized as the detection network for both the teacher and student models within the SS-OD framework. In the training stage, pseudolabels are obtained using the initialized teacher model to detect the weakly augmented unlabeled data D_{UL}^w (Fig. 2). Each pseudolabel is assigned a pseudolabel score indicating the uncertainty of the label. The low and high thresholds, τ_1 and τ_2, are employed to distinguish between the two pseudolabels. A pseudolabel is reliable, D_{PL}^R, when the pseudolabel score is greater than τ_2, and a pseudolabel is uncertain, D_L^w, when the pseudolabel score is between the low and high thresholds τ_1 and τ_2. The input data for training the student model are divided into two parts. One part is strongly augmented unlabeled data, D_{UL}^s, and the other part is weakly augmented labeled data, D_L^w, and the pseudolabels are used as the corresponding labels to calculate the loss. Therefore, the loss function L for training the student model consists of two components, the labeled loss and the unlabeled loss, as follows:

$$L = L_l + \lambda L_u \tag{1}$$

where L_l = loss function for labeled data, and L_u = loss function for unlabeled data. Parameter λ is used to balance the loss parameters of supervised learning and semisupervised learning. Loss function L_l is calculated using the following equation:

$$L_l = L_l^{\text{cls}} + L_l^{\text{reg}} + L_l^{\text{obj}} + \gamma L_{da} \tag{2}$$

where L_l^{cls} = classification loss, which calculates whether the anchor frame aligns correctly with the corresponding calibration; L_l^{reg} = localization loss, accounting for discrepancies between the predicted frame and the actual frame; and L_l^{obj} = objectness loss, reflecting the confidence level of the detected object. The labeled data loss incorporates the domain adaptation loss (L_{da}) into the original supervised learning loss function; L_{da} mitigates the neural network's overfitting to labeled data. In Eq. (2), γ is the hyperparameter for tuning L_{da}. Losses L_l^{cls}, L_l^{obj}, and L_{da} all use a binary cross-entropy loss function; L_l^{reg} uses CIoU loss, which takes into account the area of overlap, distance from the center point, and aspect ratio. Similar to L_l, L_u is determined using three types of loss functions, as follows:

$$L_u = L_u^{\text{cls}} + L_u^{\text{reg}} + L_u^{\text{obj}} \tag{3}$$

where the uncertain pseudo label determined by PLA is not involved in the loss calculation of L_u^{cls} and L_u^{reg}. The exponential moving average strategy is used to update the teacher model during the joint training process (Tarvainen and Valpola 2017). The EMA update reduces the fluctuation of model parameters due to the randomness of the training data and helps train a more stable teacher model that generates higher-quality pseudolabels. Finally, the trained student model is used as the final object detection model for real-time construction site PPE detection and violation determination.

Real-Time PPE Violation Detection

Real-time PPE violation detection utilizes the trained PPE detection model to record data by clipping PPE violation videos from construction site surveillance footage in real time, enabling safety managers to automatically obtain construction site PPE violation information. This study proposes an extraction process for PPE violation video clips (Fig. 2). First, the real-time video transmitted from the construction site camera serves as the input for the PPE detection model. Then the PPE detection results (bounding-box classes and coordinates) are input to the PPE violation determination algorithm. Construction safety managers establish PPE rules corresponding to different locations (surveillance camera IDs) and roles, based on experience and standardization. Fig. 4 details the algorithm flow for determining whether a video frame contains a violation based on the PPE detection result D and the PPE rule CR. Line 1 in the pseudocode initializes three empty sets V (PPE violation type), P (person bounding-box), and R (PPE to be worn). Lines 2–6 retrieve all person bounding-boxes and store them in the set P. Lines 9–17 traverse the CR by helmet color and camera ID to find the rules for determining the PPE to be worn. Specifically, the person bounding-box is determined by finding the helmet or head bounding-box that intersects or contains the person bounding-box. The intersecting or containing relationship between two boxes is calculated using intersection over union (IOU)

Algorithm: Real-time PPE Violation Determine

Input: PPE detection results $D = \{(c_i, b_i)\}_{i=1}^{N}$, PPE rule $CR = \{(C_j, r_j)\}_{j=1}^{M}$, camera ID d.
Output: Set of PPE violation type V.

1: Initialize V, P, R as the empty set
2: **foreach** (c, b) in D **do**
3:     **if** c is person class **then**
4:         add (c, b) to P
5:     **end if**
6: **end**
7: **foreach** (c_p, b_p) in P **do**
8:     $S \leftarrow$ Create a new set S to save bounding boxes in person bounding boxes
9:     **foreach** (c, b) in D **do**
10:         **if** IOU (b_p, b) > 0 and c is hamlet class **then**
11:             $R \leftarrow$ hamlet color and d in CR
12:         **else if** IOU (b_p, b) > 0 and c is head class **then**
13:             add no wear hamlet to V
14:         **else if** IOU (b_p, b) > 0 **then**
15:             add (c, b) to S
16:         **end if**
17:     **end**
18:     **foreach** r in R **do**
19:         **if** r not in S **then**
20:             add r to V
21:         **end if**
22:     **end**
23: **end**
24: return V

Fig. 4. Pseudocode of the proposed algorithm.

J. Constr. Eng. Manage., 2025, 151(3): 04024213

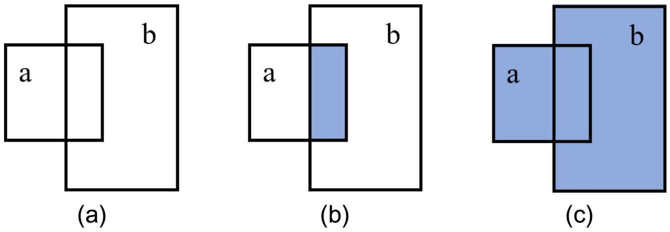

Fig. 5. Illustration of intersection over union (IOU): (a) Boxes a and b; (b) $A(a) \cap A(b)$; and (c) $A(a) \cup A(b)$.

$$\mathrm{IOU}(a, b) = \frac{A(a) \cap A(b)}{A(a) \cup A(b)} \tag{4}$$

where a and b = bounding-boxes of two respective objects [Fig. 5(a)]; $(a) \cap (b)$ denotes the area of the intersection of the two rectangular boxes a and b [Fig. 5(b)]; and $(a) \cup (b)$ denotes the union of two rectangular boxes a and b [Fig. 5(c)]. All bounding-boxes belonging to a person bounding-box can be identified by determining the IOU between the person bounding-box and other bounding-boxes. Furthermore, the role of a person bounding-box is determined by the helmet or head bounding-box. If the person bounding-box contains a helmet bounding-box, the color of the helmet and the camera ID d are used to determine the type of PPE worn by the person. If a person bounding-box contains a head bounding-box within it, then the unworn helmet is added to the violation set V. Additionally, if there are no helmets or head bounding-boxes within the person bounding-box, this indicates the inability to detect smaller PPEs, such as masks, gloves, and so forth. Under such conditions, it is not possible to determine whether there is a violation for this person, and no violation determination is made. Therefore, wrong determination due to a poor PPE detection algorithm can be avoided by this way. Line 15 in the pseudocode saves all the PPE in the person bounding-box except helmet and head in the temporary set S. Lines 18–22 of the pseudocode look for the presence of a corresponding class within the bounding-box set S based on the determination of the PPE-wearing rule R. If it does not exist, the corresponding PPE is added to violation set V. In particular, when the PPE-wearing rule R contains shoes or gloves, the PPE detection rule is to determine whether the no-shoes or no-gloves class exists.

PPE violation sequences are identified by detecting PPE violations in each video frame. Furthermore, the identified violated video is clipped using the time information of the violation video frame sequence. The time interval between two violation frames is denoted VFT. When VFT exceeds a predetermined time threshold t, this video clip is classified as a nonviolation clip, L_R. Conversely, when VFT is less than t, this video clip is classified as a suspected violation frame sequence, L_V. Subsequently, all successive sequences L_V are merged to form a single suspected violation video clip, L_P. Finally, if the number of violation frames within L_P exceeds a certain threshold ε, the suspected violation video clip is classified as a confirmed violation video clip, L_s. This distinction between a suspected and confirmed violation video clip is crucial; rapid movement or unclear focus in a frame can result in false determinations of PPE violations, leading to incorrect categorization of violation video clips. Thus, this method reduces false positives caused by erroneous detections by the PPE model, thereby enhancing the accuracy in segmenting violation videos. The confirmed violation video clips L_s are stored in the violation video database, with their corresponding violation types V. Consequently, the manager can quickly determine the current worker's status of PPE wearing at the construction site, which helps to stop the violation behavior and formulate safety management measures.

Data Set and Experiment Design

Data Set Preparation

To demonstrate the effectiveness of the proposed methodological framework, we created the construction PPE data set, CPPED. Images in the CPPED are partially sourced from the publicly available CHVG (Ferdous and Ahsan 2022) construction PPE data set. According to the method in Fig. 2, we filtered invalid images from the CHVG data set and supplemented them with a significant number of construction site PPE images sourced from site videos and the internet. Importantly, we aimed to collect a balanced number of images for each class during the selection process. A total of 2,612 valid construction PPE images were obtained, encompassing varied construction scenes, worker poses, and lighting backgrounds. Subsequently, we annotated the collected PPE images using the Labelimg (Tzutalin 2015) tool for object detection data set labeling. The total number of labeled classes was 13—helmets in 4 colors (white, blue, red, and yellow), safety vests, safety glasses, head (no helmet), persons, gloves, no-gloves (hand), shoes, no-shoes (slippers and bare feet), and masks—encompassing the 6 PPE categories in Fig. 1. During the annotation process, to ensure the accuracy of the annotated data, two graduate students completed the initial image annotation for all classes. Subsequently, a professional safety manager reviewed all the image annotations and made appropriate adjustments to the position of the annotation boxes. Finally, the CPPED data set with annotations was obtained, and its annotation files are in .txt format. Notably, four colors of helmets were used to identify four different roles of workers, and the person class was used to determine the profile of a worker. The gloves and shoes classes were not used during the violation video clips because the presence of protective gloves and safety shoes is determined by the no-gloves and no-shoes classes, respectively.

In this study, we randomly allocated 10% of the CPPED data set for testing and allocated the remaining 90% for training. The distribution of the 13 classes in both testing and training data sets is shown in Fig. 6. Among all the classes, the person class has the highest count, whereas the blue, no-shoes, and mask classes have the fewest counts. Table 1 displays the distribution of label sizes in the training and test data sets, with size division thresholds adapted from Wu et al. (2019). The CPPED data set comprises a total of 20,172 objects: 1,930 small-scale, 8,296 medium-scale, and 9,946 large-scale objects. Notably, the large-scale category contains the most objects in both the test and training data sets, with the medium-scale category trailing slightly behind. The images of large-scale objects provide more-detailed features and also facilitate better detection of small-scale objects after weak data augmentation.

Experiment Design

Three experiments were designed to verify the reliability of SS-OD methods in construction PPE detection. Experiment 1 compared the performance of the S-OD method with that of the SS-OD method at different scales of labeled data in detecting construction PPE. Concurrently, we assessed the influence of data augmentation on the performance of the S-OD method. Image data from the CPPED training data set, taken at 5% intervals from 5% to 50%, were selected as labeled data. For example, if 20% of the labeled data was selected, the remaining 80% was used as unlabeled data, exclusively for SS-OD method training. For the S-OD method, this

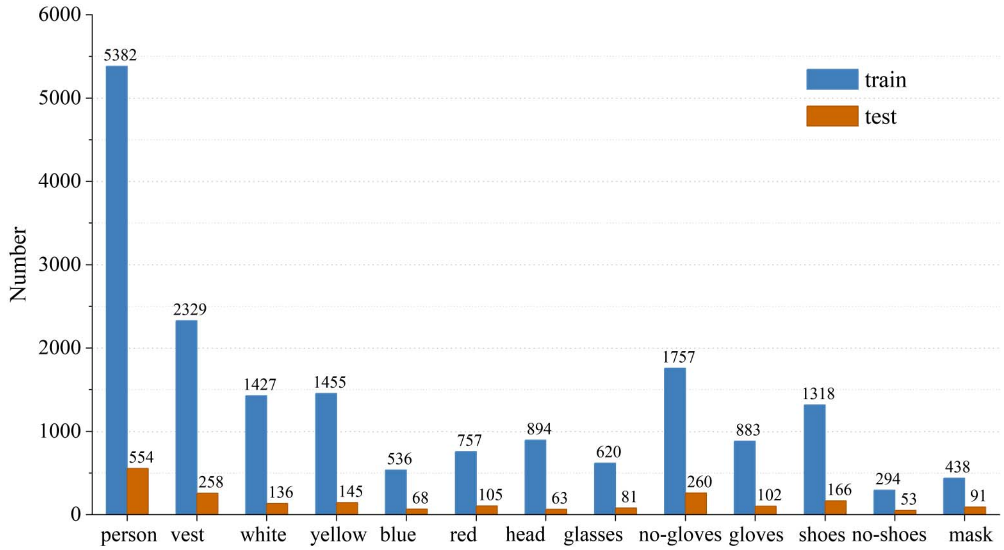

Fig. 6. Number of members of 13 classes in the test and training data sets.

Table 1. Object size distribution in training and test data sets

Data set	Small	Medium	Large	All
Testing	123	844	1,115	22,082
Training	1,807	7,452	8,831	18,090
All	1,930	8,296	9,946	20,172

Note: Small-scale: area ≤ 32^2 pixels; medium-scale: 32^2 pixels < area ≤ 96^2 pixels; and large-scale: area > 96^2 pixels.

20% of the labeled data was divided into 90% for training and 10% for validation. Experiment 2 evaluated the performance of transfer learning (TL). TL seeks to enhance the performance of an object detection model in a specific domain by leveraging knowledge from different but related source domains (Zhuang et al. 2021). This study utilized a model pretrained on the open-source Objects365 data set. The Objects365 data set, comprising over 600,000 training images across 365 object categories, includes helmets, shoes, gloves, humans, and other categories relevant to construction PPE. Hence, the performance of construction PPE detection based on the pretrained model in SS-OD methods was tested. Experiment 3 evaluated the performance of SS-OD methods for construction site PPE detection under varied scenarios. In this study, distinct scenarios were simulated by processing images from the CPPED data set to simulate foggy and low-light conditions. Fig. 7 displays both the original and the processed images (foggy and low light). The labeled data used for both S-OD and SS-OD remained unprocessed. However, the unlabeled data used for the SS-OD included all processed image data (either foggy or low light). During testing, all test images were taken from the processed data set.

In addition to the aforementioned three experiments, the performance of the proposed PPE violation video clip method was evaluated on a real construction site. First, onsite video collection was

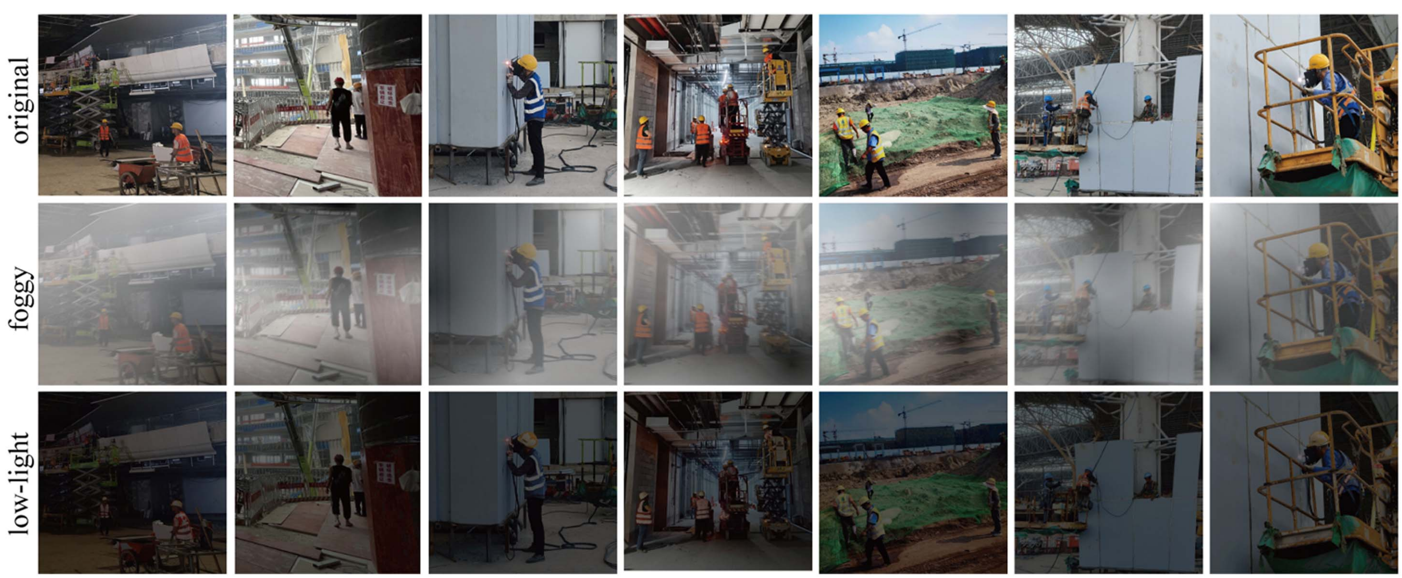

Fig. 7. Original image and images after fogging and low-light processing. (Images by Qirong Chen.)

(a) (b) (c)

Fig. 8. Video collection locations on the construction site: (a) Location A; (b) Location B; and (c) Location C. (Images by Qihua Chen.)

conducted at a selected construction project. This residential building project is located in Sichuan, China. Three distinct locations were chosen to capture real-time worker activity videos of the construction site. Each video lasted approximately 30 min, and had a resolution of $1{,}920 \times 1{,}080$ pixels and a frame rate of 30 frames/s (fps). Fig. 8 illustrates the three recording locations and their respective camera footage. Location A is situated near the entrance to the workers' living area, Location B is by the entrance of the construction access road, and Location C is at the entrance of the material handling area. Through discussions with the construction site's safety managers, we identified the categories of required PPE based on helmet color. This classification served as the basis for the PPE rules to determine if construction workers were in violation. In this construction project, workers wearing white, blue, and red helmets were required to wear safety shoes and safety vests, whereas those with yellow helmets were required to wear safety vests, safety shoes, and protective gloves. However, PPE violations within construction zones are uncommon. To introduce more instances of violations while ensuring safety, some workers were instructed to temporarily intentionally omit wearing a safety vest or protective gloves as they passed the recording locations.

Implementation Details and Evaluation Metrics

YOLOv5l was selected as the PPE detection network for both the object detection model in S-OD and the teacher and student models in SS-OD. The hyperparameters λ and γ were set to 3.0 and 0.1, respectively. After several pre-experiments, the low and high thresholds τ_1 and τ_2 for dividing the pseudolabeled categories were determined to be 0.90 and 0.97, respectively. Furthermore, the initial learning rates used for gradient descent were all set to 0.01. All models were trained on a computer equipped with an NVIDIA GeForce RTX 3090 (24 GB memory) graphics card and running the Ubuntu 20.04 operating system. During the processing of PPE violation video clips, the VFT value was set to 2 s, and the threshold ε for the number of violated frames was set to 20. Lastly, a computer with an NVIDIA GeForce RTX 3060 (6 GB memory) graphics card was employed for construction site video detection, producing both the violated video clips and the corresponding violation records.

Several evaluation metrics were utilized to comprehensively assess the aforementioned experiment: precision, recall, $F1$ score, average precision (AP), and mean average precision (mAP). Precision, which is indicative of correctness, evaluates the model's accuracy on the samples predicted as positive [Eq. (5)]. Recall evaluates the proportion of actual positive samples that the model correctly detects [Eq. (6)]. In Eqs. (5) and (6), TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively. The $F1$ score, which is a harmonic mean of precision and recall, reflects the model's capability to ensure both accuracy and completeness [Eq. (7)]. The mAP is a standard metric used to evaluate the performance of deep learning object detection algorithms, as shown in Eq. (9), where n represents the number of detected categories. The average precision accounts for both precision and recall, and is calculated as the area under the curve of the average precision at various recall rates [Eq. (8)]

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$F1 = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}} \tag{7}$$

$$\text{AP} = \int_0^1 p(r)dr \tag{8}$$

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n} AP_i \tag{9}$$

The accuracy of PPE violation clips is determined by the temporal intersection of union (TIoU) (Xiao et al. 2021a), shown in Eq. (10), where A is the manually labeled correct PPE violation clips, and B is the video clipped by the algorithm proposed in this study (Fig. 9). If the TIoU exceeds 0.5 and the PPE violation type associated with A matches that of B, then the violation clip is deemed to be correct

$$\text{TIoU} = \frac{A \cap B}{A \cup B} \tag{10}$$

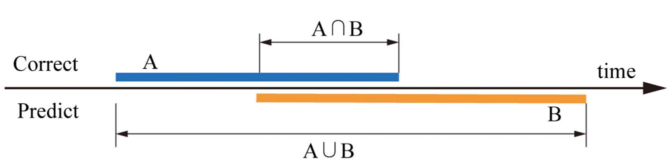

Fig. 9. Computation of TIoU.

Results

Results of Semisupervised Object Detection

According to the design of Experiment 1, the training data set from the CPPED data set was divided randomly and proportionally into labeled data. The model was trained and tested using S-OD, S-OD with weak data augmentation (WDA), and SS-OD methods. The mAP results of the three methods are shown in Fig. 10. The performance of S-OD with WDA was significantly better than that without any data augmentation. This indicates that WDA significantly improves the performance of PPE detection. The SS-OD greatly reduces the use of labeled data in achieving the same performance as S-OD. Using only 5% labeled data, the mAP for SS-OD was 63.6%, whereas it was only 49.7% for S-OD with WDA. To achieve the same results, 30% labeled data were required for the S-OD method. With 50% labeled data, the detection mAP of S-OD with WDA was only 81.9%, which did not equal the performance of the SS-OD method with 25% labeled data. In addition, the object detection performance of the S-OD method decreased in the case of 15% labeled data compared with the case of 10% labeled data. This suggests that the model's detection performance does not necessarily improve with an increase in training samples. However, the performance of the SS-OD method steadily improved with more labeled data. Therefore, the SS-OD method can achieve superior performance in PPE detection by fully utilizing unlabeled data.

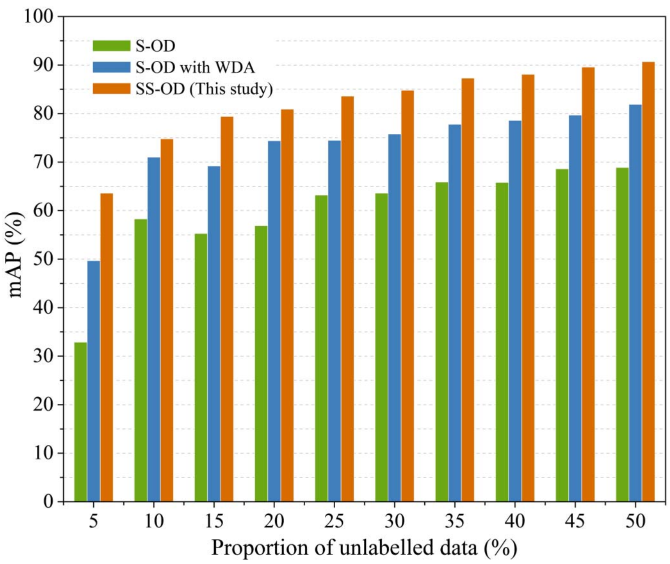

Fig. 10. Performance of S-OD, S-OD with WDA and SS-OD methods for PPE detection with different proportions of labeled data.

Table 2 presents the AP values of the 13 classes under different proportions of labeled data. With 5% labeled data, the SS-OD method achieved an AP of over 80% for seven classes (person, four colors of safety helmet, head, and mask), which are characterized by larger objects and uniform styles. Notably, the red class achieved an AP of 92%. The other five classes had lower detection performance, which is attributed largely to their smaller size and diverse styles. For example, the gloves class, which has varied styles and patterns, had an AP of only 10.8%. With 50% labeled data, these seven classes achieved AP values exceeding 95%. The mask class achieved an impressive AP of 97.5%, the highest among all classes. The performance for the classes of gloves and no-gloves was relatively low, with AP values of 67.0% and 76.6%, respectively.

Table 2 presents the performance results of Experiment 2, focusing on the PPE detection model trained with transfer learning (TL). The results reveal that SS-OD, when utilizing a pretrained model, outperformed the methods without one. Specifically, for the SS-OD method using a pretrained model, the mAP increased by 21.1%, 9.1%, and 3.5% with 5%, 25%, and 50% labeled data, respectively. This clearly demonstrates that using a pretrained model improves detection performance, especially with limited labeled data.

The results of Experiment 3 are displayed in Table 3. In both foggy and low-light construction environments, the performance of both the S-OD with WDA and SS-OD methods significantly

Table 3. Performance of S-OD with WDA and SS-OD in foggy and low-light environments

Environment	Method and labeled data proportion	Precision (%)	Recall (%)	mAP (%)
Foggy	S-OD (100%) with WDA	72.4	60.2	63.8
	SS-OD (10%)	71.0	49.0	53.4
	SS-OD (50%)	91.3	67.1	76.1
Low-light	S-OD (100%) with WDA	68.7	59.8	62.8
	SS-OD (10%)	79.5	59.6	66.6
	SS-OD (50%)	87.4	78.6	84.2

Table 2. PPE detection performance of SS-OD and SS-OD with TL methods with different proportions of labeled image

Labeled data proportion (%)	AP (%)													mAP (%)
			Safety helmet											
	Person	Vest	White	Yellow	Blue	Red	Head	Glasses	No-gloves	Gloves	Shoes	No-shoes	Mask	
5	81.3	82.4	79.5	83.5	85.7	92.0	87.2	38.6	35.1	10.8	37.1	30.7	83.3	63.6
10	87.1	89.0	87.1	89.4	92.7	95.8	92.7	63.0	50.8	26.9	51.6	55.3	90.8	74.8
15	89.0	93.4	90.3	92.8	94.1	96.3	94.1	75.5	60.7	29.5	58.1	63.9	94.4	79.4
20	92.5	92.8	91.9	91.6	94.7	97.1	94.4	70.1	58.9	38.0	66.5	71.6	92.3	80.9
25	92.1	94.3	91.8	92.5	94.1	98.3	93.5	77.4	63.3	43.9	67.8	84.7	93.3	83.6
30	92.5	94.6	92.7	94.5	93.0	98.8	97.7	77.4	65.6	47.7	67.9	83.9	96.1	84.8
35	95.0	95.9	93.4	95.1	94.8	99.2	98.3	81.9	71.1	55.1	72.7	85.7	96.6	87.3
40	95.0	96.4	93.9	95.1	95.9	98.6	99.0	84.3	71.8	57.9	73.9	87.6	95.4	88.1
45	96.1	96.3	93.7	95.5	95.6	99.1	97.0	83.1	72.4	69.3	80.8	89.7	96.5	89.6
50	96.0	96.8	92.5	95.7	97.1	98.9	98.3	88.3	76.6	67.0	80.9	93.7	97.5	90.7
5 with TL	93.7	91.7	90.1	90.1	80.9	93.7	86.9	76.4	74.0	63.6	84.3	85.2	90.8	84.7
25 with TL	96.7	92.8	94.8	94.5	97.2	99.3	95.2	85.2	85.4	81.9	88.2	95.2	98.1	92.7
50 with TL	97.4	96.3	94.8	96.8	95.0	98.1	98.5	80.2	86.2	85.9	89.1	98.8	98.1	94.2

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(a) (b) (c)

Fig. 11. PPE detection results in videos at different locations: (a) Location A; (b) Location B; and (c) Location C. (Images by Qihua Chen.)

decreased, with a more pronounced decline in the foggy environment. However, the SS-OD method with 50% labeled data outperformed the S-OD with WDA method that used 100% labeled data. Specifically, in a foggy construction environment, the SS-OD method for PPE detection achieved a detection precision of 91.3%, a recall of 67.1%, and a mAP of 76.1% using 50% labeled data. These metrics are superior to those of S-OD with WDA by 18.9%, 6.9%, and 12.3%, respectively. In a low-light construction environment, the SS-OD method for PPE detection had a detection precision of 87.4%, a recall of 78.6%, and a mAP of 84.2% with 50% labeled data, outperforming S-OD with WDA by 18.7%, 18.8%, and 21.4%, respectively. Thus, detection performance can be improved by selectively using some PPE images from specific construction environments as unlabeled data during the semisupervised training stage.

Result of PPE Violation Detection

Based on the experimental design presented in the section "Experiment Design," we used the PPE detection models trained by SS-OD (50%) to detect videos collected at construction sites. Subsequently, PPE violation video clips were extracted using the method described in the section "Real-Time PPE Violation Detection." The corresponding PPE violation type is displayed in real-time in the upper right corner of the extracted PPE violation videos, and details such as the start time of the violation clip, PPE violation type, and video name are cataloged in an Excel file. Fig. 11 illustrates the results of the PPE detection model trained by SS-OD (50%) from video clips showing violations. The PPE detection algorithm can accurately identify the PPE worn by workers on the construction site. Notably, even small objects, such as protective gloves, are detected with precision. Furthermore, the PPE detection models can discern whether the current frame contains a PPE violation based on predefined PPE rules. The primary PPE violations detected at the three locations involved the nonwearing of helmets, safety vests, and protective gloves.

Table 4 presents the detection results (in terms of precision, recall, and $F1$ score) for the violated video clips at three distinct locations using detection models trained by SS-OD (50%). The average $F1$ score for the video clips was 92.79% in the SS-OD (50%) model, demonstrating that the PPE detection model trained by SS-OD (50%) and the violated video clips performed well.

Table 4. Video clips of PPE violations performance in different positions

Position	Correct number of violation clips	Frames/s	SS-OD (50%)		
			Precision (%)	Recall (%)	$F1$ score (%)
Position A	24	61	88.46	95.83	92.0
Position B	21	57	91.30	100	95.45
Position C	17	59	93.75	88.24	90.91
Average	—	—	91.76	94.69	92.79

The average recall results for all three locations exceeded 90%, indicating that the violation video clip method was able to correctly clip almost all of the violation video clips from the construction site videos. Additionally, the computer used achieved a frame rate of approximately 60 fps for PPE detection, implying that it can process 60 video frames/s, which meets the demands of real-time processing on actual construction sites.

Discussion

Discussions of Three Experimental Results

This study presents a methodological framework for reducing the cost of the PPE detection model by using semisupervised learning and violation data recording using video clips, to address the limitations of computer vision methods in collecting PPE violation data for BBS-based construction safety risk analysis. The experimental results and findings of this study are as follows:

- Without reducing the performance of PPE detection, the demand for labeled data to train PPE detection models can be reduced using semisupervised learning, image augmentation, and transfer learning. In Experiment 1, the SS-OD method was implemented on the CPPED data set. With 50% labeled data, the mAP was 90.7% on the test data set. Using the same object detection model, YOLOv5l, the S-OD method achieved a mAP of 81.9% using 50% labeled data. Therefore, achieving higher performance with S-OD method requires more labeled data for training the PPE detection model. The performance of PPE detection models is limited by the number of labeled data, and models trained by SS-OD can improve a model's generalization ability. After applying WDA, the mAP of the S-OD method was on average 13.32% higher than that without data augmentation (Fig. 10). Consequently, data augmentation provides a broader range of data for model training and improves the performance of the trained model. In addition, the model trained by SS-OD (50%) using actual construction site demonstrated that the detection model can meet the needs of PPE detection on construction sites (Table 4). The effectiveness of transfer learning for PPE detection was tested in Experiment 2. Using the pretrained model as the initialization parameter can expedite model training and enhance the model's performance. Using transfer learning in the SS-OD method achieved a mAP value of 84.7% with 5% of labeled data, which was 21.1% higher than that without transfer learning (Table 2). However, to attain the same result, the SS-OD method without transfer learning required 30% of the labeled data. Additionally, only one open-source pretrained model was utilized in this study, suggesting that incorporating more pretrained models could optimize training further and potentially yield better PPE detection results.
- The SS-OD method adapts well to changes in construction scenarios. Construction site environments are complex and variable, making it challenging for the S-OD method to adapt to all scenarios using a fixed data set for training or to acquire more labeled data. In Experiment 3, the SS-OD method outperformed the S-OD method by 12.3% and 21.4% in the foggy and low-light construction scenarios, respectively. The SS-OD method effectively addresses the decrease in model detection accuracy caused by changes in the construction scene by integrating unlabeled data corresponding to the scene in the training process. Moreover, a significant number of unlabeled image data can be obtained easily from the construction site without the need for further labeling by experts. This not only saves time but also reduces the cost of utilizing the detection model. Actual images of the construction site or images captured in specific construction scenarios can be collected periodically, and then the PPE detection model can be trained using the SS-OD method to improve its PPE detection performance.
- The PPE violation recordings using the video clip method enhance safety management efficiency at construction sites and are applicable for BBS-based construction safety risk analysis. The method is highly flexible, and takes into account the PPE detection model capability limitations, the processing of surveillance video, and the generation of violation record data. The $F1$ score for the PPE wearing violation video clip method exceeded 90% at all three locations when using the PPE detection model trained by SS-OD (50%) (Table 4). It is evident that the PPE violation video clip method can identify the majority of PPE violations, making its deployment in real construction projects highly valuable for collecting PPE violation data and further as input data for BBS-based construction management. This approach conserves the time and effort of the safety manager and enhances the utility of surveillance footage. Concurrently, real-time assessment of PPE violations on construction sites enables safety managers to promptly address violations and avert potential hazards from escalating into accidents. Additionally, the widespread presence of surveillance cameras at construction sites encourages workers to wear proper construction PPE consistently, thereby fostering safety awareness among them.

Practical Use in Construction Safety Management

Implementing the methodology from this study at a construction site necessitates a strong infrastructure, which includes a high-quality monitoring system, a dependable network connection, and ample real-time processing capability for the servers. The process of using the proposed method in real construction projects for BBS-based safety management is shown in Fig. 12. A large number of images of workers wearing PPE are collected using surveillance cameras and smartphones at construction sites to obtain unlabeled data. The PPE detection model is trained using the SS-OD method, based on both the existing labeled and unlabeled data. The PPE detection model that yields the best training results is chosen for practical PPE detection. Then, the trained PPE detection model is deployed on the PPE violation detection platform. The platform accesses real-time surveillance videos from various locations on the construction site. The safety managers establish PPE rules based on different helmets colors (workers role) and various areas

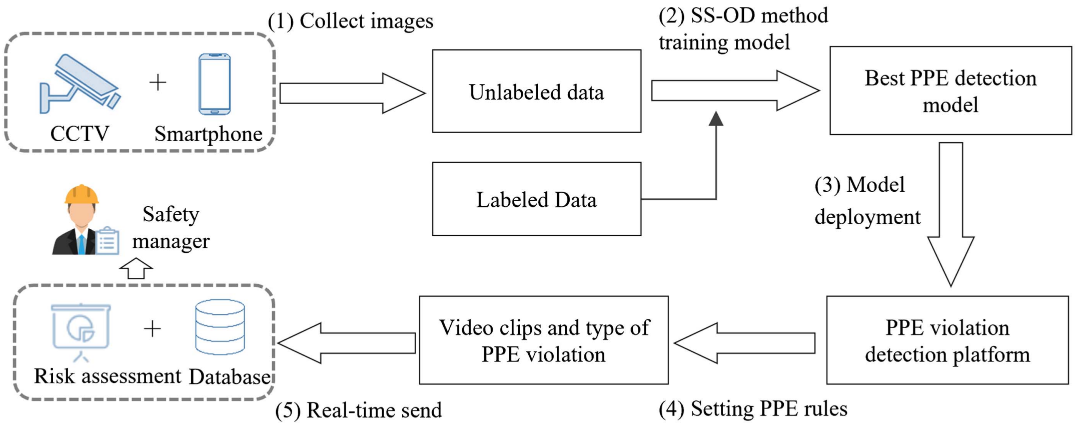

Fig. 12. Real-time PPE violation detection usage process.

(camera ID) of the construction site. PPE violation detection is performed using real-time video input from the construction site, obtaining PPE violation video clips and the corresponding violation record data. Finally, the violation video clips and recorded data, which are stored in a database, enable the implementation of appropriate safety management measures. Additionally, the resulting PPE violation records also can be utilized as input data for the assessment of unsafe behaviors in global BBS safety risk assessments on construction sites. Critical PPE violations, such as not wearing a helmet, are communicated to safety managers immediately, enabling prompt intervention in these unsafe behaviors.

The process in Fig. 12 also can accumulate substantial data on PPE violations when implemented in real construction projects. First, each surveillance camera corresponds to a unique location, enabling safety managers to identify areas with a high incidence of PPE violations. Second, these data can facilitate the automatic generation of construction safety reports. Third, safety managers can analyze PPE violation data based on different helmet colors. As a result, they can provide specific safety education for workers with high violation rates. Therefore, the process in Fig. 12 can enhance the application of behavior-based safety management. This advancement further contributes to the development of smart construction sites.

Limitations of the Proposed Method

Despite the success of the proposed method, there are some limitations to this study.

- This study did not consider the necessity of subdividing each PPE item into more-detailed detection categories, such as different types of protective gloves for varied work tasks. In addition, the imbalance of sample data in the data set is an important factor affecting the detection results. For example, the experimental results in this study show that when the number of samples in a certain category is small with the SS-OD method, the PPE detection model trained in the supervised learning stage does not generate the pseudolabels of the corresponding category effectively, resulting in poor detection of PPEs in that category.
- In this study, the worker's role was determined by helmet color and camera ID. However, the method of determining the worker's role using the color of the helmet does not permit further identification of the worker's specific identity. The integration of sensing devices and vision systems may offer an effective solution to this limitation. Thus, timely alerts to specific workers can be realized, and individual worker safety profile records can be formed.
- Although this study achieved an average accuracy exceeding 90% for video clips of PPE violations (Table 4), the locations selected for testing were limited to material handling areas and construction entrances and exits. Practical applications for construction projects of different sizes, more-complex and comprehensive construction environments, and specifications for PPE use in various construction activities, such as the use of safety ropes during aerial work, should be explored and included in this method. In addition, worker overlap, object occlusion, and small unrecognized PPEs are the primary factors causing errors in real-time PPE violation determination. Future research could address these shortcomings by integrating human posture estimation techniques and fusing information from multiple cameras.

Conclusion

Violation data collection is an important step in construction safety risk assessment in BBS-based management. This study proposes a useful framework to reduce costs through the full use of unlabeled data and collecting data on workers' PPE violations in practical construction projects using video clips. First, an SS-OD method and data augmentation are proposed to train PPE detection models. SS-OD methods can train PPE detection models using a small number of labeled data and a large number of unlabeled data. Concurrently, based on the PPE detection results, this study proposes a method for recording PPE violations using video clips from surveillance videos of construction sites. Specifically, the PPE violation detection method is employed to determine whether PPE violations occur within video frames. Sequences of video frames containing violations are utilized to extract PPE violation clips and record the types of violations, which can serve as input data for safety risk assessments, thereby improving the efficiency of construction safety management. To validate the performance of the proposed methods, a 13-class construction PPE data set (CPPED) was created for this study.

The results of the proposed SS-OD based on Yolov5 and data augmentation, and the PPE violation recording method implemented in a real construction project, showed that (1) without reducing the performance of PPE detection, the demand for labeled data in training the PPE detection model can be reduced using semisupervised learning, image augmentation, and transfer learning; (2) SS-OD methods are better equipped to handle construction scenario changes by fully utilizing unlabeled data, thereby making them suitable for construction scenarios; and (3) the violation video clips generated by the proposed method achieved an average precision of 91.76% and an average $F1$ score of 92.79% using the PPE detection model trained by the SS-OD, effectively recording the PPE violated data for BBS-based safety risk assessment.

This research makes the following contributions:

1. A construction PPE detection method using a limited number of labeled images and a single-stage object detection network is proposed by introducing a SS-OD framework. The robustness and generalization capabilities of the model trained using SS-OD were validated across various construction scenarios. Additionally, data augmentation and transfer learning techniques are integrated into the framework to enhance PPE detection performance.
2. A method for recording construction PPE violations using video clips is proposed. This method offers high flexibility, enables the automatic generation of violation records, and enhances the value of construction surveillance videos. Concurrently, real-time collected construction PPE violation data can be utilized for construction site safety risk assessments using BBS-based methods, thereby enhancing automation levels in construction safety management.

Future research will focus on expanding the types of PPE detection, enhancing model detection performance in the presence of unbalanced data samples, and advancing the introduction of sensor technology to achieve specific worker identification. Additionally, to address PPE violations in specific construction tasks—such as the need for safety glasses during operations such as drilling, cutting, and sanding—we will endeavor to improve existing methods to enhance PPE violation detection.

Data Availability Statement

Some or all data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request. The CPPED data set created by this study can be downloaded at https://drive.google.com/drive/folders/1pd2_kzUA82M25s8HVL6rHCUUWCaTdVnv?usp=sharing.

Acknowledgments

The authors acknowledge the financial support provided by the National Key Research and Development Program of China (Grant Nos. 2022YFC3801104 and 2021YFB2600501).

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